

Antiderivatives and Indefinite Integrals

ANSWER KEY



Basic antiderivatives

- 1 Find each indefinite integral:

a $\int (3x^2 - 4x + 5) dx = x^3 - 2x^2 + 5x + C$

b $\int \frac{1}{x^2} dx = -\frac{1}{x} + C$

c $\int \sqrt{x} dx = \frac{2}{3}x^{3/2} + C$

d $\int \sec^2 x dx = \tan x + C$

- 2 Find $\int (4\sin x + e^x) dx$.
 $= -4\cos x + e^x + C$.

Using initial conditions

- 3 Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.
 $f(x) = 2x^3 - 2x + C$. Using $f(1) = 5$: $2 - 2 + C = 5$, so $C = 5$: $f(x) = 2x^3 - 2x + 5$.
- 4 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.
 $s(t) = \int v(t) dt = t^3 - 3t^2 + C$. Using $s(0) = 4$: $C = 4$: $s(t) = t^3 - 3t^2 + 4$.
- 5 An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s².
a Find the velocity function $v(t)$. $v(t) = -9.8t$ m/s (using $v(0) = 0$).
b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$. $s(t) = -4.9t^2 + 45$.
c Find how long it takes to hit the ground ($s = 0$), to two decimal places. $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

More antiderivatives

- 6 Find each indefinite integral:
- a** $\int (\frac{1}{x} - 3x^{-4}) dx = \ln|x| + x^{-3} + C$
- b** $\int (2\sec x \tan x) dx = 2\sec x + C$
- c** $\int \frac{4}{\sqrt{1-x^2}} dx = 4\sin^{-1} x + C$
- 7 Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).
Treating $(x + 1)$ as a single unit (power rule for a linear inner function): $\int (x + 1)^3 dx = \frac{(x + 1)^4}{4} + C$.

Motion applications

8 A rocket's acceleration is $a(t) = 6t \text{ m/s}^2$, with $v(0) = 0$ and $s(0) = 0$.

a Find $v(t)$. $v(t) = 3t^2$.

b Find $s(t)$. $s(t) = t^3$.

c Find the distance traveled in the first 4 seconds. $s(4) = 64 \text{ m}$.

Antiderivatives of composite expressions

9 Find $\int (2x - 1)^5 dx$, treating $(2x - 1)$ as a linear inner function.

$$\int (2x - 1)^5 dx = \frac{(2x - 1)^6}{6 \cdot 2} + C = \frac{(2x - 1)^6}{12} + C \text{ (dividing by the derivative of the inner function, 2).}$$

10 Find $\int \sin(3x) dx$ and $\int e^{5x} dx$.

$$\int \sin(3x) dx = -\frac{1}{3} \cos(3x) + C. \quad \int e^{5x} dx = \frac{1}{5} e^{5x} + C.$$

Verifying antiderivatives by differentiation

11 Verify that $F(x) = x \ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.

$$F'(x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x = f(x) \checkmark.$$

Antiderivatives of reciprocal and rational-power functions

12 Find each indefinite integral:

a $\int \frac{3}{x} dx = 3 \ln|x| + C$

b $\int x^{-2/3} dx = 3x^{1/3} + C$

c $\int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$

A projectile motion application

13 A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8 \text{ m/s}^2$. Find its velocity function $v(t)$ and position function $s(t)$.

$$v(t) = \int a(t) dt = -9.8t + C_1. \text{ Using } v(0) = 15: v(t) = -9.8t + 15.$$

$$s(t) = \int v(t) dt = -4.9t^2 + 15t + C_2. \text{ Using } s(0) = 2: s(t) = -4.9t^2 + 15t + 2.$$

Visualizing an antiderivative family

- 14** The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F .
- a** Write the general antiderivative of $f(x) = 2x$. $F(x) = x^2 + C$.
- b** If a specific antiderivative passes through $(1, 4)$, find its constant C . $1^2 + C = 4$, so $C = 3$: $F(x) = x^2 + 3$.
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Antiderivatives requiring algebraic rewriting first

- 15** Find $\int \frac{x^3 + 2x}{x} dx$ by simplifying the integrand before integrating.
 $\frac{x^3 + 2x}{x} = x^2 + 2$, so $\int (x^2 + 2) dx = \frac{x^3}{3} + 2x + C$.
- 16** Find $\int (x + 1)(x - 3) dx$ by expanding the product first.
 $(x + 1)(x - 3) = x^2 - 2x - 3$, so $\int (x^2 - 2x - 3) dx = \frac{x^3}{3} - x^2 - 3x + C$.